LUND UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written Examination
Group and Ring Theory
MATP33
Saturday, 30 November 2024
Duration: 8:00–13 :00

No books, notes, computational devices, etc. are allowed. Use clear handwriting and give clear careful motivations. Please write your anonymous identification number on each sheet of paper.

1. Show that every group of order 30 has a non-trivial normal subgroup.

2. Determine the rational canonical form of

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}.$$

considered as a real matrix.

3. Which of the following groups of order 300 are isomorphic:

$$\mathbb{Z}_{300};\ \mathbb{Z}_{10} \oplus \mathbb{Z}_{30};\ \mathbb{Z}_5 \oplus \mathbb{Z}_6 \oplus \mathbb{Z}_{10};\ \mathbb{Z}_{15} \oplus \mathbb{Z}_{20};\ \mathbb{Z}_2 \oplus \mathbb{Z}_{150}?$$

4. Let $R = M_n(\mathbb{C})$ be a ring of complex matrices of size n, and I be a non-zero left ideal in R. Show that I, considered as a left $R-$module is simple if and only if all non-zero matrices in I have rank 1.

5. Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, B = \begin{pmatrix} 5 & 6 \\ 7 & a \end{pmatrix}$$

be two real matrices. Considering them as linear maps $A, B : \mathbb{R}^2 \to \mathbb{R}^2$ find the matrix that corresponds to the linear map $C = A \bigotimes_{\mathbb{R}} B$. For which a the map C is invertible?

6. Let R be a left Noetherinag ring with unity and $ab = 1$ for some elements $a, b \in R$. Show that $ba = 1$ as well.